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Thermodynamic Study of Er-Bi and Er-Te Systems by Combination of First-Principles Calculations and the CALPHAD Method

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Abstract The CALPHAD method combined with first-principles calculations was used to estimate the thermodynamic properties of the intermediate phases in the Er-Bi and Er-Te systems and thermodynamic descriptions of the phase diagrams of these systems. The enthalpies of formation of the $ErBi_2$, ErBi, Er_5Bi_3, $ErTe_3$, Er_2Te_3, and ErTe compounds at 0 K were calculated by first-principles method to supply the thermochemical data for this optimization. The associated solution model was used to describe the Gibbs energy of the liquid phase of both systems, while the intermetallic phases were modeled as stoichiometric compounds. A set of self-consistent thermodynamic parameters of the Er-Bi and Er-Te systems is obtained.

Keywords CALPHAD method · Er-Bi and Er-Te systems · first-principles calculations · thermodynamic assessment

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1 Introduction

Rare earth elements and their alloys are being intensively studied since they are widely used in new technologies, optoelectronics, medicine, automotive, liquid-phase catalysts, metallurgy, digital technologies, and are especially indispensable in the development of environmentally friendly energy.[1–6]

Some thermodynamic modeling of bismuthides of rare earth elements can be found in the literature.[7–12] Thus, the development of new alloys from Er-Bi and Er-Te systems using computational thermodynamics requires an accurate thermodynamic description of these binary systems using CALPHAD (Calculation of Phase Diagrams) technique.[13] This method makes it possible to relate phase diagrams and thermodynamic properties, and it is a powerful tool for reducing the cost and time of material design.[14,15]

In this work, the enthalpies of formation of the intermetallic compounds in the Er-Bi and Er-Te systems were determined from first-principles calculations before the thermodynamic optimization in order to overcome the scattering of thermodynamic data. The Er-Te system is optimized for the first time and Er-Bi is reassessed. The Er-Bi system was previously evaluated twice using the subregular solution model.[16,17] These two estimates did not take into account the compound $ErBi_2$, although the existence of this compound was experimentally confirmed by Abdusalyamova and Rakhmatov.[18] In addition, an allotropic transformation of the ErBi compound was reported,[18] which was not taken into account in the two previous optimizations.

2 Literature Review

2.1 The Er-Bi System

The phase diagram of the Er-Bi system, given in the Massalski compilation,[19] was originally proposed by Moffatt[20] based on the literature data.[21–24] Okamoto[25] presented a phase diagram constructed by Abulkhaev[26] based on thermal analysis, metallographic and X-ray methods. Only two stoichiometric compounds have been identified: ErBi (cubic, NaCl type) and Er_5Bi_3 (orthorhombic,Y_5Bi_3 type).[22,23,27] It is expected the Er_5Bi_3 bismuthide peritectically decomposes at 1700 K into liquid and solid ErBi phases. The highest melting phase in the system belongs to ErBi, which melts congruently at 2080 K. The solubility of bismuth in erbium is less than 1 at.% Bi. Later, Abdusalyamova and Rakhmatov[18] re-examined the phase diagram of the Er-Bi system using DTA, XRD and metallography, and they discovered a new stoichiometric compound $ErBi_2$, that decomposes peritectically at 868 K. The $ErBi_2$ phase was taken into consideration in this work, although its crystal structure and lattice parameters are still unknown. The Er_5Bi_3 phase is formed by peritectic reaction at 1700 K. It was reported that there are eutectic points at 1423 K and 542 K for the compositions 19.5 at.% Bi, and 99.7 at.% Bi, respectively. The ErBi compound has a solid-state phase transformation at 2053 K, [18] but the crystallographic structure of this phase at high temperature has not been determined. Both phase diagrams redrawn from Okamoto[25,28] are shown in Fig. 1. The crystal structure and lattice parameters of pure elements and the three compounds (α-ErBi, $ErBi_2$, and Er_5Bi_3) are shown in Table 1.[16,21,29–31]

Furthermore, Wang et al.[16] optimized the binary system, based on the Abulkhaev[26] phase diagram with only two compounds ErBi and Er_5Bi_3 and a compilation of structural data of the intermetallic compounds that were determined by Yoshihara et al.[21] The terminal solid solutions of the Er-Bi system were described without homogeneity ranges. Later Wang et al.[17] performed a remodeling of the same phase diagram based also on the experimental data of Abulkhaev[26] taking into consideration the terminal solid solution (Er) but without the new compound $ErBi_2$ and the allotropic transformation of the ErBi compound given by Abdusalyamova and Rakhmatov.[18]

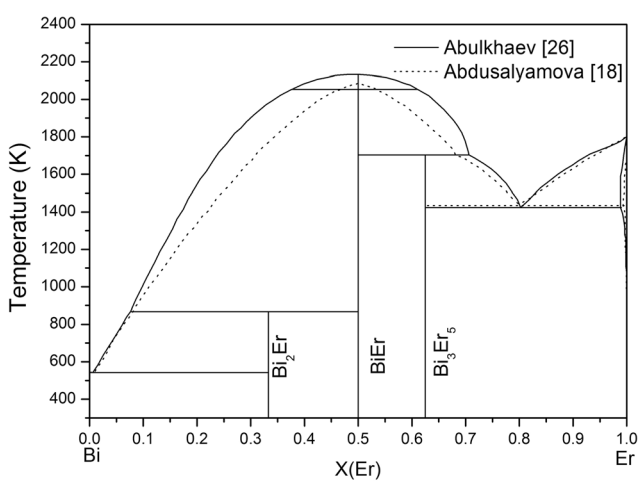

Fig. 1 Experimental phase diagrams of the Er-Bi systems reviewed by Okamoto[25,28]

The enthalpies of formation of the compounds ErBi and Er_5Bi_3 in the temperature range from 725 K to 1065 K have been determined by Yamshchikov[32] and Petrashkevich[33] using EMF method. Besides, Borzone et al.[34] and Sidorko[35] measured again the enthalpy of formation of the ErBi intermetallic at 300 K and in the range 805-1093K, using direct synthesis calorimetry and EMF method respectively. Colinet et al.[36], using the Miedema model, predicted that the value of the enthalpy of mixing of the liquid phase for 50 at.% Bi is -43 kJ/mol-at (reference states are pure liquid metals). The same enthalpy has been calculated by Takeuchi and Inoue[37] using the Miedema model with modified parameters.

2.2 The Er-Te System

The Er-Te system was studied by Haase et al.[38] using thermal analysis and X-ray diffraction. The elemental Er of 99.9% purity and Te of 99.99% purity were pre-reacted in Vycor tubes at 600 °C for 48 hr. The reaction temperature was limited up to 700 °C because at temperatures more than 700 °C the erbium reacted with Vycor. The pre-reacted samples with compositions greater than 40 at.% Er were then sealed in tantalum tubes in an argon environment and heated to temperatures of approximately 1800 °C. According to this work, three compounds have been detected. The ErTe and Er_2Te_3 compounds had congruently melting temperatures at 1773 and 1486 K, respectively. While the $ErTe_3$compound melts at 957 K with decomposition according to the peritectic reaction: $Er_2Te_3 + L \leftrightarrow ErTe_3$. The ErTe and Er_2Te_3 compounds formed a continuous series of solid solution between these stoichiometries. The eutectic between Er and ErTe is given at 87 at.% Er and 1551 K, while the second eutectic on the Te-rich side is at about 3 at.% Er and 603 K (Fig. 2). Later Abrikosov et al.[39] reinvestigated the Er-Te phase diagram and confirmed the existence of ErTe, Er_2Te_3 and $ErTe_2$ phases. A degenerate eutectic exists between $ErTe_2$ and Te at 718 K. According to reference,[39] the $ErTe_3$ phase does not exist; instead, the stoichiometric phase $ErTe_2$, formed by a peritectic at 953 K, is indicated. The liquidus line in the region with less than 40 at.% Er determined by

Table 1 The crystal structure data of the Er-Bi and Er-Te systems

Phase	Pearson symbol	Space group	Lattice parameters (nm)			Method	Refs.
			a	b	c		
Bi	*hR*6	$R\bar{3}m$	0.4555	0.4555	1.2024	Exp.	29
			0.4556	0.4556	1.2025	DFT	16
			0.4582	0.4582	1.2170	DFT	This work
$ErBi_2$	*mP*12	$P12_1/c1$	0.4180	0.7608	1.1495	DFT	This work
ErBi(α)	*cF*8	$Fm\bar{3}m$	0.62023	0.62023	0.6202	Exp.	21
			0.6264	0.6264	0.6264	DFT	16
			0.6279	0.6279	0.6279	DFT	This work
Er_5Bi_3	*oP*32	*Pnma*	0.8090	0.9349	1.1806	Exp.	30
			0.8171	0.9446	1.1889	DFT	16
			0.8190	0.9457	1.1919	DFT	This work
Er	*hP*2	$P6_3/mmc$	0.3495	0.3495	0.5486	Exp.	31
			0.3513	0.3513	0.5517	DFT	16
			0.3581	0.3581	0.5553	DFT	This work
$ErTe_3$	*oC*16	*Cmcm*	0.4285	0.4285	2.5278	Exp.	40
			0.4325	0.4325	2.5384	DFT	This work
Er_2Te_3	*oF*80	*Fddd*	1.2127	0.8581	2.5728	Exp.	43
			1.2031	0.8565	2.4908	DFT	This work
ErTe	*cF*8	$Fm\bar{3}m$	0.6063	0.6063	0.6063	Exp.	44
			0.6109	0. 6109	0. 6109	DFT	This work
$ErTe_2$	*tP*6	*P4/nmm*	0.4248	0.4248	0.8865	Exp.	45*
			0.4448	0.4448	0.8691	DFT	This work
Te	*hP*3	$P3_121$	0.4440	0.4440	0.5920	Exp.	46
			0.4513	0.4513	0.5961	DFT	This work

*The synthesis of the $ErTe_2$ compound was carried out at high pressures (up to 100 kbar) and temperatures (up to 1473 °C).

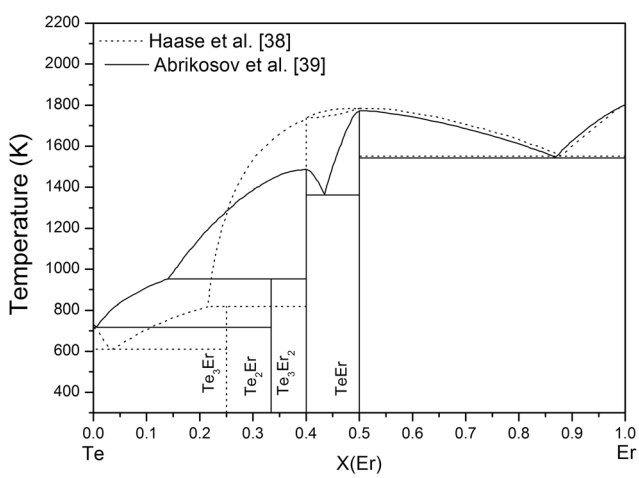

Fig. 2 Experimental phase diagrams of the Te-Er systems reviewed by Vassiliev et al.[40]

Abrikosov et al.[39] is more credible than that given by Haase et al.[38] In the work[38] the alloys with less than 40 at.% Er were heated only at 600 °C, however, the alloys with more than 40 at.% Er were heated up to 1800 °C.

The information of the Er-Te phase diagram system was compiled by Vassiliev et al.[40] and Hinz et al.[41] Vassiliev et al.[40] studied the Er-Te system in the range from 60 to 100 at.% Te, using EMF method and X-ray diffraction. Their results confirmed the existence of the two compounds Er_2Te_3 and $ErTe_3$. The $ErTe_3$ phase was confirmed by Zhang et al.[42] using X-ray analysis. The Er-Te phase diagram given in the Massalski handbook[19] is redrawn from the results Abrikosov et al.[39], but the compound $ErTe_3$ was taken into consideration instead $ErTe_2$.

Brixner[47] and Khan and Garcia[44] reported that ErTe intermetallic has an fcc, NaCl- type structure. Stoewe[43] takes the Er_2Te_3 compound as a slightly deformed cubic structure of the Sc_2S_3-type. The compound $ErTe_3$ has an orthorhombic structure and melts incongruently.[40] The structure of the $ErTe_{2-x}$ phase was determined by Canon and Hall[45], where the synthesis of this phase was carried out at high pressures (up to 100 kbar) and temperatures (up to 1473 °C). According to these, authors, the synthesis of the $ErTe_{2-x}$ compound is highly improbable at the standard temperature and pressure. We have not found any information about the crystallographic structure of the $ErTe_{2-x}$

compound at the standard conditions. Crystal structure and lattice parameters of the pure elements and the binary compounds are summarized in Table 1.[40,43–46] Thermodynamic data of the Er-Te system were studied by the EMF measurements.[40,48–51]

3 First-Principles Calculation

The thermochemical information for the intermetallic compounds available in the literature is rather uncertain, as the scatter between the measured data is considerable. Thus, first-principles calculations based on density functional theory[52] were employed to calculate the enthalpies of formation of the binary compounds in the Er-Bi and Er-Te systems, and test the constancy of the experimental data. The Vienna ab-initio simulation package (VASP)[53,54] was used with plane wave basis sets. The calculations were performed using the Project Augmented Wave (PAW) pseudo-potential[55,56] and employed the Generalized Gradient Approximation (GGA) of Perdew, Burke and Ernzerhof (PBE).[57]

The energy of formation of the compounds at 0K were described by an equation of the following form:

$$\Delta E\left(Er_pA_q\right) = E\left(Er_pA_q\right) - \frac{p}{p+q}E(Er) - \frac{q}{p+q}E(A), (\mathrm{A} = \mathrm{Bi\ \ or\ \ Te}) \qquad \text{(Eq 1)}$$

Where E(Er), E(A), and E(Er_pA_q) are pure hcp_A3 (Er), pure rho_A7 (Bi), pure hex_A8 (Te) and the total energy of the Er_pA_q compound at 0 K respectively.

Since no phase transitions occur for the pure elements and compounds between 0 K and room temperature, the enthalpy of formation of the binary compounds can be approximated to the energy of formation of the compounds at 0 K[58,59]

4 Thermodynamic Modeling

4.1 Unary Phases

The expression ${}^0G_i^{\varphi}(T) = G_i^{\varphi}(T) - H_i^{SER}(298.15K)$ represents the molar Gibbs energy for pure element i (i=Bi, Te or Er) in the phase φ (φ=liquid, Er (hcp_A3), Bi (rho_A7) and Te (hex_A8)), with the molar enthalpy H_i^{SER} for its stable element reference (SER) state. This expression may be written as:

$${}^0G_i^{\varphi}(T) = a + bT + cTLnT + dT^2 + eT^3 + fT^{-1} + gT^7 + hT^{-9} \qquad \text{(Eq 2)}$$

Where T is the absolute temperature, and a, b, c, d, e, f, g and h are the temperature coefficients. In our investigation, the Gibbs energy of each pure element was obtained from the SGTE compilation.[60]

4.2 Solution Phases

4.2.1 Substitutional Solution Model

The substitution model was used only to describe the primary solid solution (Er) in the Er-Te system. The primary solid solutions, for the rest of the elements, were taken to be negligible. The Gibbs energy of phase φ is the sum of three terms: reference, ideal and excess term:

$$G^{\varphi} = G_{\varphi}^{ref} + G_{\varphi}^{ide} + G_{\varphi}^{exc} = x_{Er}G_{Er}^{\varphi} + x_AG_A^{\varphi} + RT[x_{Er}Ln(x_{Er}) + x_ALn(x_A)] + G_{\varphi}^{exc} \qquad \text{(Eq 3)}$$

Where R is the gas constant, x_i is the mole fraction of element i in the φ phase and T is the temperature. The excess Gibbs energy is described using the Redlich-Kister polynomial model[61] and given by the following expression:

$$G_{\varphi}^{exc} = x_{Er}x_A\sum_{\vartheta=0}^{n}{}^{\vartheta}L_{\varphi}(x_{Er} - x_A)^{\vartheta} \qquad \text{(Eq 4)}$$

Where ${}^{\vartheta}L_{\varphi}$ (ϑ = 0, 1, 2 . . .) are the interaction parameters that have been taken as temperature dependent and can be written as ${}^{\vartheta}L_{\varphi} = a_v + b_vT$. a_v and b_v are the parameters to be optimized.

4.2.2 Associate Model

The associated solution model proposed by Sommer[62] usually gives a better physical description of the short-range ordering behavior in the liquid phase than the sub-regular solution model, thus yielding more reliable parameters.

The relatively high melting points of the ErBi compound in the Er-Bi system and the asymmetric shape of the liquidus around the ErTe compound in the Er-Te system indicate the existence of short-range order in the liquid phase. Moreover, the presence of short-range order in the liquid phase was noted in several Bi and Te rare-earth binary systems.[7,63–65] Therefore, the associated solution model has been adopted to describe the thermodynamic properties of the liquid phases in Er-Bi and Er-Te systems. For the Er-Bi system, the ErBi associate is included in the model. In the case of the Er-Te system, using only ErTe as an associate, we failed to achieve the existing eutectic

Table 2 The standard enthalpies of formation of the intermetallic compounds in the Er-Bi and Er-Te systems

System	Compound	ΔH_f (kJ/mol-at.)	Method	References
Er-Bi	$ErBi_2$	−45.237	DFT	This work
		−53.528	CALPHAD	This work
		−59.000	Miedema model	[36]
	α-ErBi	−80.005	DFT	This work
		−79.131	CALPHAD	This work
		−80.274	DFT	[64]
		−77.576	DFT	[65]
		−82.069	DFT	[16]
		−83.620	CALPHAD	[16]
		−80.350	CALPHAD	[17]
		−103.00	EMF	[32]
		−109.00	EMF	[33]
		−90.000	Calorimetry	[34]
		−101.30	EMF	[35]
		−75.000	Miedema model	[36]
	Er_5Bi_3	−63.496	DFT	This work
		−62.917	CALPHAD	This work
		−64.084	DFT	[64]
		−62.157	DFT	[65]
		−66.035	DFT	[16]
		−70.280	CALPHAD	[16]
		−66.486	CALPHAD	[17]
		−69.000	Miedema model	[36]
Er-Te	$ErTe_3$	−84.589	DFT	This work
		−86.964	CALPHAD	This work
		−84.996	DFT	[64]
		−116.258	DFT	[65]
		−102.85±1.3	EMF	[50]
	Er_2Te_3	−133.731	DFT	This work
		−136.475	CALPHAD	This work
		−133.951	DFT	[64]
		−157.965	DFT	[65]
		−159.1±10	Predicted	[48]
		−164.44±16.7	EMF	[40]
		−158.62±2.5	EMF	[50]
	ErTe	−115.927	DFT	This work
		−118.947	CALPHAD	This work
		−119.399	DFT	[64]
		−136.485	DFT	[65]
		−179.45±0.6	EMF	[50]
		−157.00±60	EMF	[51]
	$ErTe_2$	−105.846	DFT	This work
		−106.3902	DFT	[64]
		−132.921	DFT	[65]

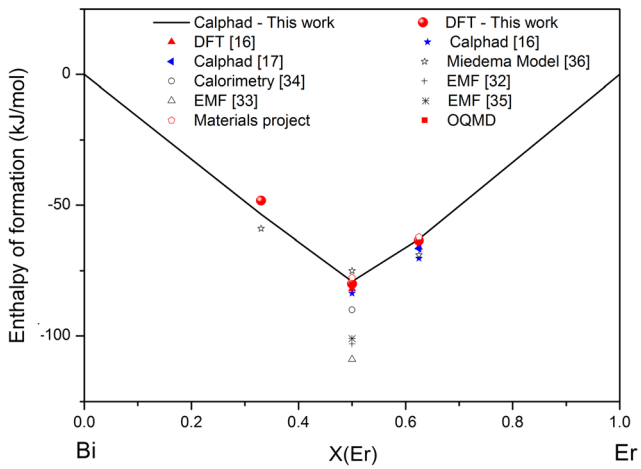

Fig. 3 Standard enthalpies of formation of intermediate phases in the Er-Bi system

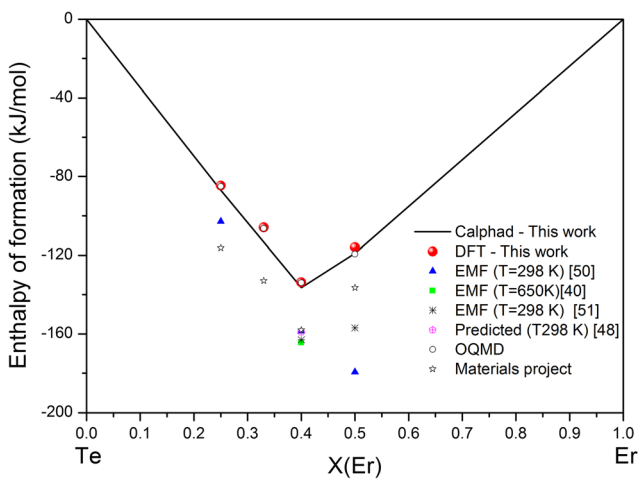

Fig.4 Standard enthalpies of formation of intermediate phases in the Er-Te system

between ErTe and Er_2Te_3. For this reason, the ErTe and Er_2Te_3 associates were adopted for the Er-Te system.

The Gibbs energy of the liquid is described by the following equation:

$$G^{Liq} = x_{Er}{}^{0}G_{Er}^{Liq} + \sum_i x_i{}^{0}G_i^{Liq} + RT\left[x_{Er}Ln(x_{Er}) + \sum_i x_i Ln(x_i)\right] + G_{Liq}^{exc} \quad \text{(Eq 5)}$$

Where, G_{Liq}^{exc} and $x_i$are the excess Gibbs energy and the mole fraction of each species i (i = Bi and ErBi for Er-Bi system, and Te, Er_2Te_3 and ErTe for Er-Te system). The excess Gibbs energy is expressed by this formula.

Table 3 Calculated enthalpies of formation of $ErBi_2$ for different suspected structures

Pearson Symbol	Space Group Number	Prototype	ΔH_f, kJ/mol-at
*hP*3	164	CdI_2	39.6679
*hP*12	143	MoS_2	15.2241
*cP*12	205	FeS_2	−15.4151
*oS*6	21	$ErSb_2$	−16.5698
*oP*12	19	Ag_2Se	−21.1173
*cF*12	225	$CoSi_2$	−23.1992
*hP*6	194	MoS_2	−23.4701
*hP*9	150	Fe_2P	−25.1344
*oP*6	58	FeS_2	−25.4140
*tI*6	139	$MoSi_2$	−33.6478
*mC*6	12	$AuTe_2$	−35.9422
*tI*6	140	Al_2Cu	−38.4406
*tP*6	129	$ErTe_2$	−43.2532
*mS*48	9	$LaAs_2$	−43.6048
*oP*12	62	Co_2Si	−45.1803
*oS*12	63	$ErSn_2$	−45.2102
*mP*6	11	$CaSb_2$	−45.9664
*oS*24	64	$SmSb_2$	−47.7533
*mP*12	14	$NdAs_2$	−48.2365

$$G_{Liq}^{exc} = \sum_i \sum_{j\neq i} x_i x_j \sum_{\vartheta=0}^{n} {}^{\vartheta}L_{i,j}^{Liq}\left(x_i - x_j\right)^{\vartheta} \quad \text{(Eq 6)}$$

Where i and j correspond to the species: Bi, ErBi and Er (Te, Er_2Te_3, ErTe and Er) in the Er-Bi (Er-Te) systems. ${}^{\vartheta}L_{i,j}^{Liq}$ is the ϑth ordered interaction parameter between the species i and j, which is a function of temperature as: ${}^{\vartheta}L_{i,j}^{Liq} = {}^{\vartheta}a_{i,j} + {}^{\vartheta}b_{i,j}T$.

4.3 Intermetallic Compounds

The intermetallic compounds in the Er-Bi and Er-Te binary systems were all considered as stoichiometric phases and their Gibbs energy of formation was determined with the formula:

$$\Delta_f G^{Er_pA_q} = \frac{p}{p+q}{}^{0}G_{Er} + \frac{q}{p+q}{}^{0}G_A + \Delta_f H^{Er_pA_q} - T\Delta_f S^{Er_pA_q} \quad \text{(Eq 7)}$$

Where ${}^{0}G_{Er}$ and ${}^{0}G_A$ denote the Gibbs energy of the pure elements Er and A (A: Bi or Te) respectively. $\Delta_f H^{Er_pA_q}$ and $\Delta_f S^{Er_pA_q}$ are the enthalpy and entropy of formation of the Er_pA_q compound, to be calculated in the present optimization.

Table 4 The optimized parameters of the Er-Bi system

Phases	Thermodynamic Parameters (Gibbs energy in J/mol, temperature in Kelvin)
Liquid	${}^0G^{Liq}_{ErBi} = {}^0G^{Liq}_{Bi} + {}^0G^{Liq}_{Er} - 91635 - 9.86T$
Formula(Bi, ErBi,Er)	${}^0L^{Liq}_{Bi,ErBi} = -64159 + 12.11T$
	${}^0L^{Liq}_{ErBi,Er} = -68995 + 11.59T$
	${}^1L^{Liq}_{ErBi,Er} = -14351 + 1.48T$
(Er) Formula(Er,Bi)	${}^0L^{(Er)}_{Er,Bi} = -138109 + 17.12T$
$ErBi_2$	$\Delta_f G^{ErBi_2} = -160586 - 5.91T + 2GHSERBI + GHSERER$
α-ErBi	$\Delta_f G^{\alpha ErBi} = -158263 + 0.32T + GHSERBI + GHSERER$
β-ErBi	$\Delta_f G^{\beta ErBi} = -156174 - 0.70T + GHSERBI + GHSERER$
Er_5Bi_3	$\Delta_f G^{Er_5Bi_3} = -503336 - 2.0T + 3GHSERBI + 5GHSERER$

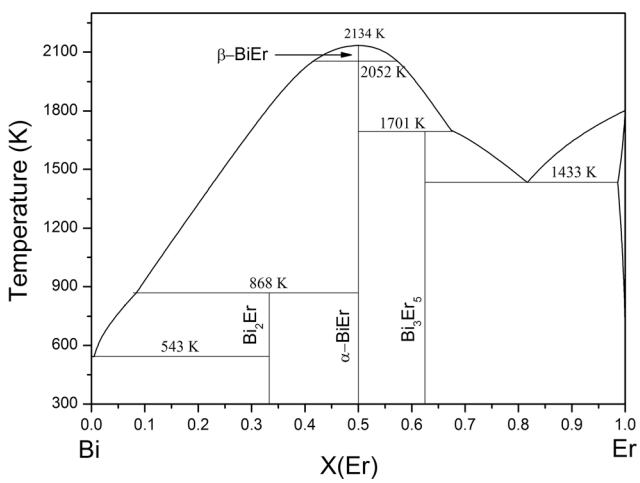

Fig. 5 Our calculated phase diagram of the Er-Bi system.

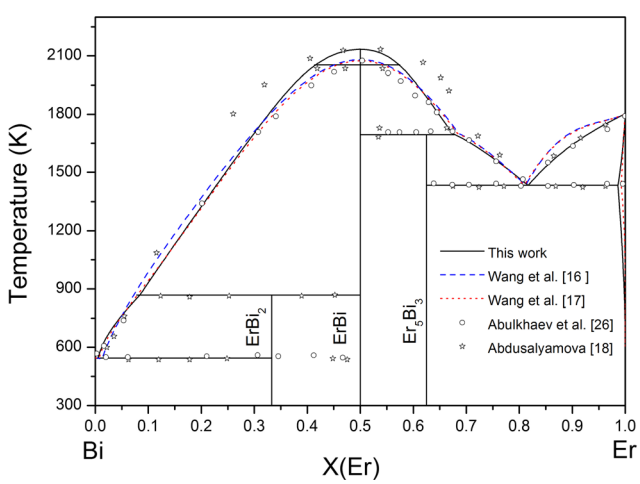

Fig. 6 Calculated Er-Bi phase diagram in comparison with the experimental and assessed data.

5 Results and Discussion

5.1 First-Principles Calculations

First, we have studied the known structures of compounds in the two systems. A plane wave cutoff energy of 500 eV is used for all compounds, and K-point meshes are compiled using the fully automatic scheme.[63] Convergence on the number of K-points in the irreducible wedge of the first Brillouin zone is checked for each structure.

The calculated lattice parameters reported in Table 1 are in good agreement with experimental data, with a maximum difference of about 3 %. The calculated enthalpies of formation of the compounds in the two systems are given in Table 2 and Figs. 3, 4, along with the experimental data and predictions from the first-principles calculations taken from the OQMD[64] and Materials Project[65] online databases.

The existence of the intermetallic $ErBi_2$ has been reported by Abdusalyamova and Rachmatov[18] but without structural characteristics. Since the crystal structure of the $ErBi_2$ compound is unknown, we searched in the AFLOW online database[66] for the possible structures in which this compound can be crystallized, and we have found 19 crystal structures with A_2B stoichiometrie. We have calculated the structural parameters of the relaxed geometries and the enthalpies of formation of each structure. The crystal structures and calculated enthalpies of formation of all evaluated structures are summarized in Table 3.

The proposed structure for this compound is *mP*12 structure (space group $P12_1/c1$) which has the lowest enthalpy of formation. This enthalpy is about 4 kJ/mol-at above the ground state line (Fig. 3), so this phase may not

Table 5 The invariant reactions in the Er-Bi system including calculated and experimental results

Invariant reaction	Type of reaction	Compositions of the respective phases (x_{Er})			T,K	Method	Reference
L ↔(Bi) + $ErBi_2$	Eutectic	0.005	0	0.333	543	CALPHAD	This work
		0.003	0	0.333	542	Exp.	18
L+ ErBi ↔ $ErBi_2$	Peritectic	0.08	0.5	0.333	868	CALPHAD	This work
		0.08	0.5	0.333	868	Exp.	18
L ↔ ErBi	Congruent	–	0.5	–	2134	CALPHAD	This work
		–	0.5	–	2080	Exp.	26
		–	0.5	–	2133	Exp.	18
		–	0.5	–	2080	CALPHAD	16
			0.5	–	2074	CALPHAD	17
β-ErBi ↔ α-ErBi	Allotropic	—	0.5	—	2052	CALPHAD	This work
		—	0.5	—	2053	Exp.	18
L+ ErBi ↔ Er_5Bi_3	Peritectic	0.67	0.5	0.625	1701	CALPHAD	This work
		0.68	0.5	0.625	1700	Exp.	26
		0.71	0.5	0.625	1700	Exp.	18
		0.68	0.5	0.625	1700	CALPHAD	16
		0.68	0.5	0.625	1700	CALPHAD	17
L ↔ Er_5Bi_3+(Er)	Eutectic	0.82	0.625	0.985	1433	CALPHAD	This work
		0.80	0.625	> 0.99	1433	Exp.	26
		0.80	0.625	0.98	1423	Exp.	18
		0.81	0.625	1	1434	CALPHAD	16
		0.81	0.625	0.99	1434	CALPHAD	17

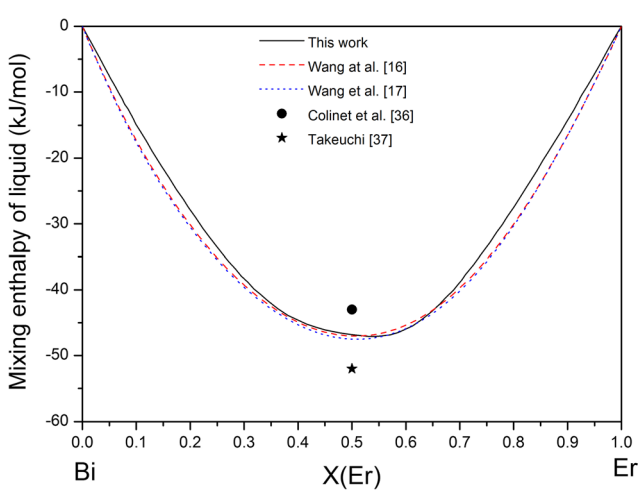

Fig. 7 Mixing enthalpies of the liquid phase in the Er-Bi system.

be stable at 0 K or it will have another structure. The existence of this compound will be studied during the thermodynamic optimization of the Er-Bi system, where the effect of temperature will be considered with the Gibbs energy of formation. Moreover, the thermodynamic optimization by the CALPHAD method improves the enthalpies of formation compared to those calculated by first-principles calculations at 0K.

For the Er_2Te compound, synthesized by Canon and Hall[45] at high pressure and temperature, we performed calculations based on the structure they proposed. As seen from Fig. 4, the $ErTe_2$ compound does not break the convex hull. The difference between the calculated enthalpy of formation and the convex hull is 8 kJ / mol-at. The $ErTe_2$ compound also cannot be stable at high temperatures. This is confirmed by the experimental results of Vassiliev et al.[40] and Zhang et al.[42] Thus, we did not consider the $ErTe_2$ compound during thermodynamic optimization.

5.2 Thermodynamic Assessment

The thermodynamic optimization of the Gibbs energy parameters is part of the CALPHAD approach.[14,15] It was carried out by using the PARROT module in the Thermo-Calc software version 2018a.[67] A set of optimized model parameters are obtained, each experimental datum is weighted carefully according to its merits, and changed by trial and error during the assessment, reaching a good agreement with the experimental data within the expected uncertainty limits.

Table 6 The optimized parameters of the Er-Te system

Phases	Thermodynamic Parameters (Gibbs energy in J/mol, temperature in Kelvin)
Liquid phase, Formula(Te, ErTe, Er_2Te_3,Te)	${}^0G^{Liq}_{ErTe} = {}^0G^{Liq}_{Te} + {}^0G^{Liq}_{Er} - 219890 + 13.95T$
	${}^0G^{Liq}_{Er_2Te_3} = 3{}^0G^{Liq}_{Te} + 2{}^0G^{Liq}_{Er} - 428300 - 23.65T$
	${}^0L^{Liq}_{Er,Te} = -185312 + 18.87T$
	${}^1L^{Liq}_{Er,Te} = -49412 + 2.63T$
	${}^0L^{Liq}_{ErTe,Er} = -12430 + 9.98T$
	${}^1L^{Liq}_{ErTe,Er} = 11242 - 3.95T$
	${}^0L^{Liq}_{Te,Er_2Te_3} = -120254 - 19.5T$
	${}^0L^{Liq}_{Er_2Te_3,ErTe} = -118412 - 8.22T$
	${}^1L^{Liq}_{Er_2Te_3,ErTe} = -35512$
$ErTe_3$	$\Delta_f G^{ErTe_3} = -347856 + 36.32T + 3GHSERTE + GHSERER$
Er_2Te_3	$\Delta_f G^{Er_2Te_3} = -682375 + 78.275T + 3GHSERTE + 2GHSERER$
ErTe	$\Delta_f G^{ErTe} = -237894 + 3.356T + GHSERTE + GHSERER$

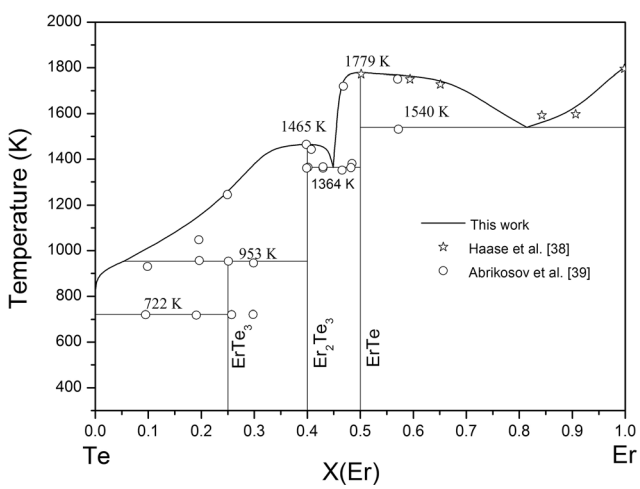

Fig. 8 Our calculated Er-Te phase diagram compared with the experimental data.

5.2.1 *The Er-Bi System*

The parameters of the thermodynamic description of the Er-Bi system obtained in the present assessment are given in Table 4. Calculated Er-Bi phase diagram is shown in Figure 5 and compared with the experimental data of Abulkhaev[26], Abdusalyamova and Rachmatov[18] and assessed[16,17] in Fig. 6. The estimated phase diagram agrees better with experimental data than previous calculations,[16,17] especially in the Er-rich part. The liquidus line in the region adjacent to the ErBi phase is lower than the experimental data of Abdusalyamova and Rakhmatov,[18] although we used their experimental data. The intermetallic phase $ErBi_2$ proposed by Abdusalyamova and Rachmatov[18] was confirmed with a peritectic decomposition at 868 K. The invariant reactions in the Er-Bi system are summarized in Table 5, where the experimental data and calculated ones are also listed for comparison. The difference between the calculated and the measured invariant equilibrium temperature of Abdusalyamova and Rachmatov[18] is less than 5 K.

Figure 7 shows the calculated enthalpies of the liquid phase with two predicted values at 50 at.% Er obtained by using the Miedema model.[36,37] As we can see, our results are acceptable if we consider the large uncertainties of the predicted values. The calculated enthalpies of formation from CALPHAD modeling are shown in Fig. 3 with these obtained from first-principles calculations and the literature data.[16,17,32–36,64,65] The enthalpies of formation of the ErBi and Er_5Bi_3 compounds obtained from the CALPHAD optimization and first-principles calculations agree well with the previous assessment and the experimental data from the EMF method. For the new compound $ErBi_2$, the optimized enthalpy of formation is close to the one predicted by Colinet et al.[36] The allotropic transformation enthalpy of the ErBi compound was found to be 1.044 kJ/mol-at.

5.2.2 *The Er-Te System*

The experimental data of the phase diagram of Haase et al.[38] and Abrikosov et al.[39] made the thermodynamic optimization of this system very difficult. According to the first-principles calculations, where we have found that the

Table 7 The invariant reactions in the Er-Te system including experimental data

Invariant reaction	Type of reaction	Compositions of the respective phases (x_{Er})			T,K	Method	References
L↔ (Te) +$ErTe_3$	Eutectic	9.1E-7	0.25	0	722	CALPHAD	This work
		0.0044	0.25	0	718	Exp.	39
		0.0044	0.25	0	718	Review	19
L+ Er_2Te_3↔ $ErTe_3$	Peritectic	0.05	0.40	0.25	953	CALPHAD	This work
		0.22	0.40	0.25	848	Exp.	38
		0.14	0.40	0.25	953	Review	19
L ↔ Er_2Te_3	Congruent	–	0.40	–	1465	CALPHAD	This work
		–	0.40	–	1463	Exp.	39
		–	0.40	–	1486	Review	19
L ↔ Er_2Te_3+ErTe	Eutectic	0.45	0.40	0.50	1364	CALPHAD	This work
		0.44	0.40	0.50	1363	Exp.	39
		0.44	0.40	0.50	1363	Review	19
L ↔ ErTe	Congruent	–	0.50	–	1779	CALPHAD	This work
		–	0.50	–	1763	Exp.	39
		–	0.50	–	1773	Exp.	38
		–	0.50	–	1773	Review	19
L ↔ ErTe+(Er)	Eutectic	0.82	0.50	1	1540	CALPHAD	This work
		0.87	0.50	1	1543	Exp.	38,39
		0.87	0.50	1	1543	Review	19

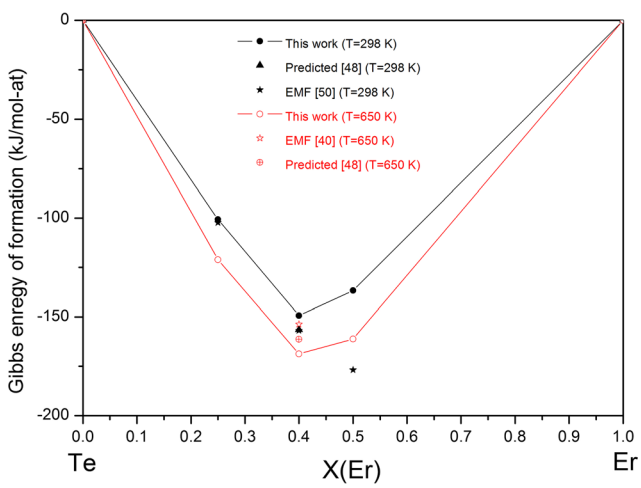

Fig.9 Standard Gibbs energy of formation of intermediate phases in the Er-Te system.

$ErTe_3$ compound is energetically stable unlike the $ErTe_2$ intermetallic, we have ignored the $ErTe_2$ compound in the assessment. For the liquid phase we have initially used only ErTe as an associate since it has the highest melting temperature, but in all the attempts we made we failed to reproduce the eutectic invariant between ErTe and Er_2Te_3. For this reason, the ErTe and Er_2Te_3 associates were used with Er and Te to describe the excess Gibbs energy of the liquid phase with the associate model.

The parameters of the thermodynamic description of the Er-Te system obtained in the present assessment are given in Table 6. Figure 8 shows the calculated Er-Te phase diagram using the present set of thermodynamic parameters in comparison with the experimental data, [38,39] where we can see that the calculated liquidus line is very close to the experimental data. The calculated invariant reaction temperatures and compositions together with the experimental one are listed in Table 7. The $ErTe_3$ tellurium-rich phase was confirmed. Its peritectic decomposition temperature is in good agreement with that reported in the Massalski compilation.[19] This temperature is higher than the results of Haase et al.[38] The previously reported $ErTe_2$ by Abrikosov et al.[39] is not confirmed.

Figure 4 shows the evaluated enthalpies of formation of intermetallic phases determined by the CALPHAD method together with the calculated values determined by first-principles calculations, as well as the experimental data obtained by the EMF method[40,48,50,51] and taken from the OQMD and Materials Project online databases.[64,65] A satisfactory agreement was obtained between the optimized value and those calculated by first-principles calculations in this work and from OQMD database.[64] The data from Materials Project[65] are slightly lower since it was modified with anion correction for Te element. The enthalpies of formation determined indirectly by the EMF method[40,48,50,51] are lower than those calculated. Our calculations are acceptable if we take into account the large uncertainties of the EMF method, especially in the determination of the enthalpies of formation. In addition, in the

work of Imamaliyeva, [50] the alloys of this system are not well annealed and measurements by the EMF method in glycerol give values outside of thermodynamic equilibrium. The calculated Gibbs energy of the compounds at 298 K and 650 K are shown in figure 9 and compared with the experimental data. As shown in this figure, a satisfactory agreement is obtained unless the ErTe compound where the values of Imamaliyeva[50] are lower than the calculated ones. The thermodynamic stability of the ErTe phase increases with increasing temperature relative to the other compounds.

6 Conclusion

The Er-Bi and Er-Te systems have been studied by combination of first-principles calculations and the CALPHAD approach. The Er-Te system has been assessed for the first time, and the Er-Bi system has been reassessed. Due to the inconsistency between the experimental data, in particular the compounds existing in each system, we carried out a first-principles study of all the compounds proposed to determine their stabilities.

The assessment of the two systems has been performed by using the experimental phase diagrams data and enthalpies of formation of the intermetallic phases obtained by first-principles calculations. A consistent set of thermodynamic parameters from various thermodynamic models has been obtained, for each solution phase and intermetallic compounds in the Er-Bi and Er-Te binary systems. The thermodynamic stability of the $ErBi_2$ and $ErTe_3$ phases has been confirmed. According to the results obtained in this work, we strongly suggest that the continuation of the experimental investigation is necessary for the Er-Bi and Er-Te systems to confirm the inferences deduced, particularly in the bismuth and tellurium rich parts.

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